

Module 6. Cluster Analysis and Selected Methods

- ❑ Cluster Analysis: Basic Concepts
- ❑ Partitioning Methods 
- ❑ Hierarchical Methods
- ❑ Density-based and Grid-based Methods
- ❑ Advanced Methods
- ❑ Evaluation of Clustering
- ❑ Summary

Partitioning Algorithms: Basic Concepts

- Partitioning method: Discovering the groupings in the data by optimizing a specific objective function and iteratively improving the quality of partitions
- K-partitioning method: Partitioning a dataset D of n objects into a set of K clusters so that an objective function is optimized (e.g., the sum of squared distances is minimized, where c_k is the centroid or medoid of cluster C_k)

- A typical objective function: Sum of Squared Errors (SSE)

$$SSE(C) = \sum_{k=1}^K \sum_{x_i \in C_k} |x_i - c_k|^2$$

- Problem definition: Given K , find a partition of K clusters that optimizes the chosen partitioning criterion
 - Global optimal: Needs to exhaustively enumerate all partitions
 - Heuristic methods (i.e., greedy algorithms): K-Means, K-Medians, K-Medoids, etc.

The K-Means Clustering Method

- ❑ K-Means (MacQueen'67, Lloyd'57/'82)
 - ❑ Each cluster is represented by the center of the cluster
- ❑ Given K, the number of clusters, the K-Means clustering algorithm is outlined as follows
 - ❑ Select K points (may not be the actual data points) as initial centroids
 - ❑ Repeat
 - ❑ Form K clusters by assigning each point to its closest centroid
 - ❑ Re-compute the centroids (i.e., mean point) of each cluster
 - ❑ Until convergence criterion is satisfied
- ❑ Different kinds of measures can be used
 - ❑ Manhattan distance (L1 norm), Euclidean distance (L2 norm), Cosine similarity